

Disclaimer

- This training resource was developed to be used with the CZ ID platform
- In October, 2026, stewardship of CZ ID transitioned to UCSF and the platform was renamed to Seqtold
- These training materials are not being actively maintained and may not reflect differences between CZ ID and Seqtold, subsequent changes to the platform, or current methods and best practices in metagenomic sequencing and analysis



Generating heatmaps using ONT data from CZ ID

Web-app to create heatmaps

- CZ ID does not have a heatmap visualization feature for the ONT pipeline
- [This web app](#) is designed to mimic this functionality using sample taxon reports downloaded from CZID ([Github link](#))

Step 1: Download sample taxon reports

The screenshot shows the iD CZ ID web interface. A modal dialog titled "Select a Download Type" is open, indicating that 36 samples are selected. The dialog lists four report types under the heading "REPORTS":

- ☐ **Sample Metadata** (sample_metadata.csv)
User-uploaded metadata, including sample collection location, collection date, sample type
- ☒ **Sample Taxon Reports** (.csv)
Computed metrics (e.g. total bases, bPM) and metadata for each taxon identified in the sample
- ☐ **Combined Sample Taxon Results** (.csv)
The value of a particular metric (e.g. total bases, bPM) for all taxa in all selected samples, combined into a single file
- ☐ **Samples Overview** (.csv)
Sample QC metrics (e.g. percent reads passing QC) and other summary statistics

A blue button labeled "Start Generating Download" is at the bottom of the modal. Below the button, it states: "Downloads for larger files can take multiple hours to generate."

In the background, the interface shows a list of samples. The first visible sample is "RLB11_control1_LongA..." with a status of "COMPLETE", dated "2026-06-11", from "Human" in "Bolivia", with "1,978" reads and "77.43%" coverage. The left sidebar contains filters for Taxon (TAXON, ANNOTATION) and Metadata (LOCATION, TIMEFRAME, VISIBILITY, HOST, SAMPLE TYPE). The right sidebar shows summary statistics for the selected samples, including "OVERALL" metrics like "Samples: 59", "Projects: 1", "Avg. reads per sample: 12,134,366", and "Avg. reads passing filters per sample: 1,409,879".

Step 2: Open the app and upload reports

[Link to web app](#)

Heatmap Parameters

Upload CZID Sample Taxon Reports (CSV) ?

 Upload

200MB per file • CSV

Filter Settings

Metric to display ?

nt_bpm ▼

Taxonomic Level ?

Species × × ▼

Categories ?

viruses × × ▼

CZID Nanopore Heatmap Generator

Upload your CZID sample taxon reports to generate an interactive heatmap. Adjust parameters in the sidebar to customize your visualization.

 [Upload CZID sample taxon report CSV files using the sidebar to get started](#)

Instructions

How to use this app:

1. **Download your data from CZID:**
 - Use the CZ ID bulk download workflow to download Sample Taxon Reports
 - Navigate to https://czid.org/bulk_downloads
 - Download and unzip the CSV reports
2. **Upload your files:**
 - Use the file uploader in the sidebar
 - Select multiple CSV files from your CZID download



Heatmap shows abundance with threshold options on the left panel

Taxonomic Level

Species

Categories

viruses

Threshold Filters

Min NT bpm

10.00

Min NR bpm

0.00

Min reads (r)

0

Min NT Contigs

0



Optionally select samples to include in a background model

The screenshot shows a web interface with a sidebar on the left and a main content area on the right. The sidebar contains three sections: 'Taxonomic Level' with a 'Species' tag, 'Categories' with a 'viruses' tag, and 'Threshold Filters' with three input fields: 'Min NT bPM' (10.00), 'Min NR bPM' (0.00), and 'Min reads (r)' (0). The main content area has a 'Fork' button and a green success message: '✓ 10 file(s) uploaded successfully'. Below this is a button 'View uploaded files'. A section titled 'Background Model (Optional)' contains the text 'Select background samples to calculate z-scores for anomaly detection.' and a dropdown menu 'Select Background Samples' with the option 'Choose options' selected. A blue callout box next to the dropdown says 'Select background samples to enable z-score filtering'.

Taxonomic Level ?

Species × ▼

Categories ?

viruses × ▼

Threshold Filters

Min NT bPM ?

10.00 − +

Min NR bPM ?

0.00 − +

Min reads (r) ?

0 − +

Fork 🔄 ⋮

✓ 10 file(s) uploaded successfully

> View uploaded files

Background Model (Optional)

Select background samples to calculate z-scores for anomaly detection.

Select Background Samples ?

Choose options ▼

Select background samples to enable z-score filtering

The background model uses the same Z-score calculation as the Standard model used in the CZ ID Illumina pipeline, except using bPM as the abundance metric instead of rPM

Set Z-score threshold for use in the background model

The screenshot shows a web interface with two main panels. The left panel contains settings for taxonomic level, categories, and threshold filters. The right panel shows a 'Background Model (Optional)' section with a list of background samples and a Z-score threshold slider.

Taxonomic Level ⓘ
Species ×

Categories ⓘ
viruses ×

Threshold Filters

Min NT bPM ⓘ
10.00 - +

Min NR bPM ⓘ
0.00 - +

Min reads (r) ⓘ
0 - +

Background Model (Optional)

Select background samples to calculate z-scores for anomaly detection.

Select Background Samples ⓘ

- RLB02_Water_NP_filtered ×
- RLB05_Water_NP_filtered ×
- RLB05_Water_Serum_filtered ×
- RLB02_Water_Serum_filtered ×

Min Z-score Threshold ⓘ
1.00 - +

Choice of Z-score threshold depends on the levels of contamination observed, but a good starting point is a threshold of 1.00

Heatmap can be download as .png image file or .csv data file

nc_001101

Taxonomic Level

Species ×

Categories

viruses ×

Threshold Filters

Min NT bPM

10.00 - +

Min NR bPM

0.00 - +

Min reads (r)

0 - +

Min NT Contigs

0 - +

